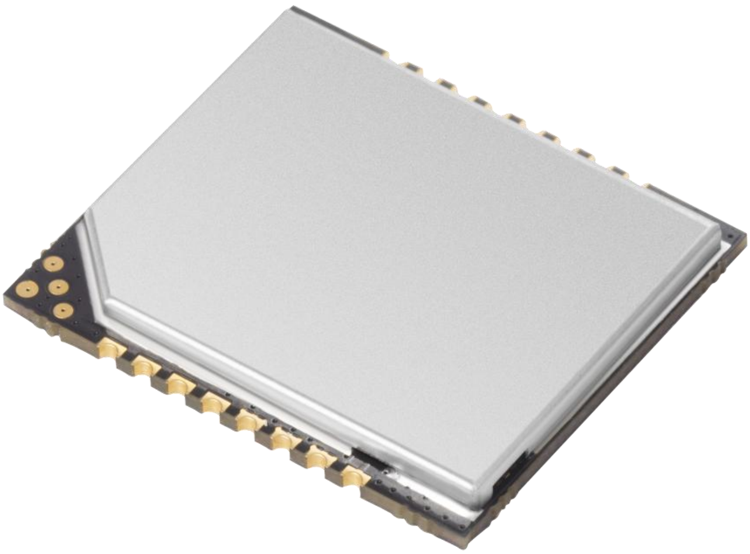
YR300

Micro UHF RFID Module



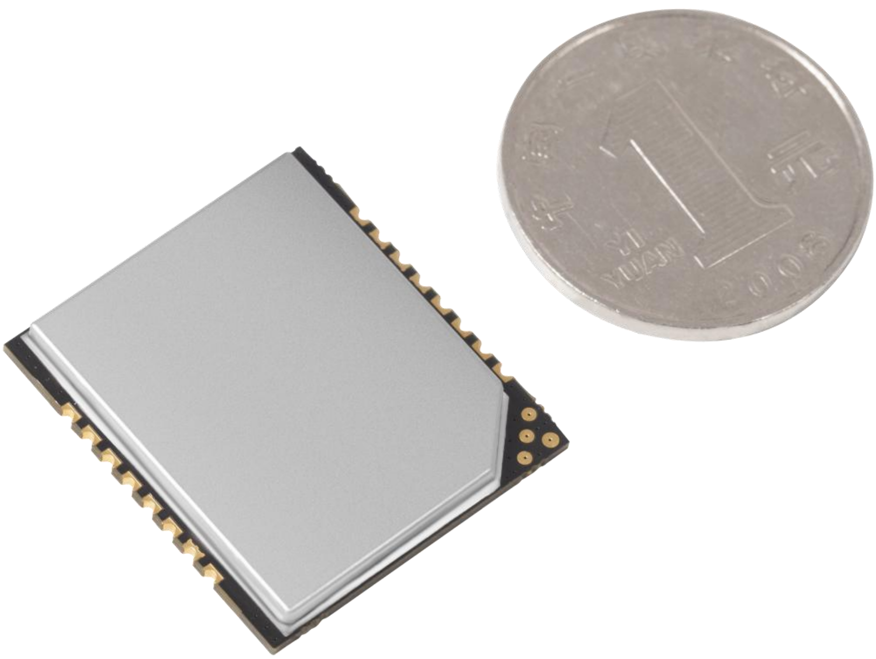
Extremely compact

Supreme performance

1. Advantages

- High sensitivity up to -85 dBm.
- Extremely compact design and excellent performance bring superior cost-effectiveness.
- Low power consumption, no need to add external heat sinks.
- Very easy to be embedded (Refer to circuit design Figure 5-1).

2. Product View



3. PIN Assignments

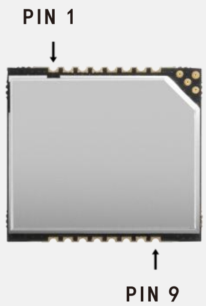


Figure 3-1

PIN	Interface	Description
1	Antenna	Antenna connection (Impedance 50Ω)
2	GND	Grounding
3	GND	
4	GND	
5	GND	
6	GND	
7	GPIO 2	Input GPIO
8	GPIO 1	Input GPIO
9	VCC	Power supply DC 5V
10	GND	Grounding
11	EN	High level enable
12	RXD	UART RXD
13	TXD	UART TXD
14	Beeper	Connect the buzzer (Driven externally is recommended)
15	GPIO 3	Output GPIO
16	GPIO 4	Output GPIO

4. Electrical Parameters

Electrical Parameters				
Input Voltage	DC 5V			
Standby Mode Current	＜35mA（EN high level)			
Sleep Mode Current	＜100μA（EN low level)			
Operating Current	Conditions	Min	Typical	Max
	@5V(25dBm Output，25°C)	300mA	320mA	380mA
Operating Temperature	-20°C~+70°C			
Storage Temperature	-20°C~+85°C			
Work Humidity	5%RH~95%RH(non-condensing)			
Protocol	EPC global UHF Class 1 Gen 2 / ISO 18000-6C			
Spectrum Range	902 ~ 928MHz, 865 ~ 868MHz Optional			
Supported Regions	US, Canada and other regions following U.S. FCC Europe and other regions following ETSI EN 302 208 China, Korea, Malaysia			
RF Output Power	0 ~ 25dBm			
Output Power Precision	+/- 1dB			
Output Power Flatness	+/- 0.2dB			
Receive Sensitivity	< -85 dBm			
Peak Inventory Speed	> 50 tags/sec			
Tag RSSI	Supported			
Communication Interface	Uart 3.3V			
GPIO	2 inputs 2 outputs (3.3V TTL Level)			
Serial Baud Rate	115200 bps（Default and Recommended）/38400 bps			
Dimensions	28 * 25 * 2.7mm			
Heat Dissipation	Air cooling			

5. Circuit Design

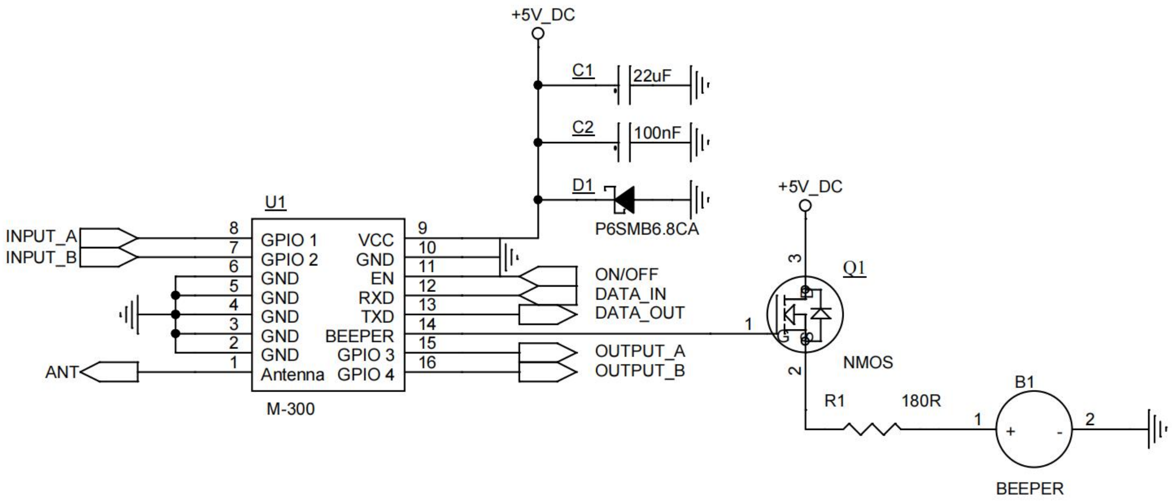
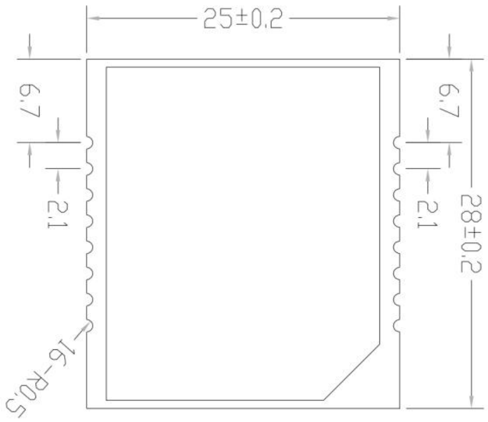


Figure 5-1 Circuit Design For Reference

6. Dimensions (Unit：mm)

Note: If there is inconsistency between the image and the actual product, please refer to real product as standard.



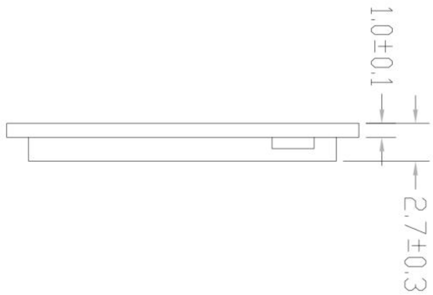


Figure 6-1 Product dimensions

Figure 6-2 Product thickness